LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Practice Pack: Work and Energy

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

For teacher use only. Do not distribute to students.

LEVEL 1: CONCEPTS

Q1. Define work and give its SI unit.

Answer: Work is done when a force moves an object in the direction of the force: W = F × s. The SI unit is the joule (J). 1 J = 1 N × 1 m.

Q2. Give two situations where a force acts but no work is done.

Answer: (1) Pushing a wall that does not move (no displacement). (2) Carrying a load on your head while walking on level ground: the force of gravity is perpendicular to the displacement, so gravity does no work.

Q3. Define power and give its SI unit.

Answer: Power is the rate of doing work (P = W/t). The SI unit is the watt (W) = 1 J/s.

Q4. State the law of conservation of energy.

Answer: Energy can neither be created nor destroyed. It can only change from one form to another, so the total energy stays the same.

Q5. How many joules are there in 1 kWh?

Answer: 1 kWh = 1000 W × 3600 s = 3.6×10^6 J.

LEVEL 2: NUMERICALS

Q6. A force of 7 N moves an object 8 m in the direction of the force. How much work is done?

Answer: W = F × s = 7 × 8 = 56 J.

Q7. Find the kinetic energy of a 15 kg object moving at 4 m/s.

Answer: KE = ½mv^2 = ½ × 15 × 4^2 = 120 J.

Q8. Find the potential energy of a 10 kg object at a height of 6 m. (g = 9.8 m/s^2)

Answer: PE = mgh = 10 × 9.8 × 6 = 588 J.

Q9. A lamp uses 1000 J of electrical energy in 10 s. What is its power?

Answer: P = W/t = 1000/10 = 100 W.

Q10. How much work is needed to increase the speed of a 1500 kg car from 30 km/h to 60 km/h?

Answer: u = 30/3.6 ≈ 8.33 m/s, v = 60/3.6 ≈ 16.67 m/s. W = change in KE = ½$m(v^2 - u^2)$ = ½ × 1500 × (277.8 – 69.4) ≈ 1.56×10^5 J (156,250 J).

Q11. A 1500 W heater runs for 10 hours. How much energy does it use, in kWh?

Answer: E = P × t = 1.5 kW × 10 h = 15 kWh.

LEVEL 3: CHALLENGE (FOUNDATION)

Q12. Two girls, each weighing 400 N, climb a rope to a height of 8 m. Girl A takes 20 s and girl B takes 50 s. Find the power of each.

Answer: Work by each = 400 × 8 = 3200 J. Power of A = 3200/20 = 160 W. Power of B = 3200/50 = 64 W.

Q13. A boy of mass 50 kg runs up a staircase of 45 steps in 9 s. Each step is 15 cm high. Find his power. ($g = 10\ m/s^2$)

Answer: Height = 45 × 0.15 = 6.75 m. Work = mgh = 50 × 10 × 6.75 = 3375 J. Power = 3375/9 = 375 W.

Q14. An object of mass 40 kg is raised to a height of 5 m. Find its potential energy. If it is allowed to fall, find its kinetic energy when it is halfway…

Answer: PE = mgh = 40 × 10 × 5 = 2000 J. Halfway down, PE = 40 × 10 × 2.5 = 1000 J. By conservation of energy, KE = 2000 – 1000 = 1000 J.